

Computer science Mini NEA

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Contents

1	Analysis	2
1.1	Problem Description	2
1.2	Background	2
1.3	Research	2
1.4	Interview	2
1.4.1	Resulting Objectives	4
1.4.2	Extended Objectives (if i have time after completing initial objective) . .	4
1.5	Analysis of Existing Solutions	4
1.6	Objectives	4
1.7	Modelling	4
2	Documented design	5
2.1	Overall flowchart	5
2.2	Hierarchy diagram	5
3	Technical solution	7
4	Testing	8
5	Evaluation	9

1 Analysis

1.1 Problem Description

Physics teachers often desire a effective method of demonstrating the principles of projectile motion they teach in lessons in a visual manner that makes it easy for students to understand how they can be applied in real life.

1.2 Background

Projectiles are specifically defined as any object thrown or launched into the air that moves solely under the influence of gravity. The motion of a projectile in the air is relatively easy to calculate using given formula's derived from Newton's initial XUVAT equations. However, the overall motion can be hard to visualise and so it can often be difficult to see how the parameters interact to form the equations, resulting in only surface level knowledge. A tool that automatically calculates and displays projectile motion would help students to consolidate this knowledge and thus create a more concrete understanding of the topics the teacher is trying to explain.

1.3 Research

Equations that could be used in calculations

1. finding horizontal and vertical velocity components at any point in time using the initial velocity and angle from the horizontal

$$v_x = v_0 \cos(\theta) \quad (1)$$

$$v_y = v_0 \sin(\theta) - gt \quad (2)$$

2. velocity at any point in time using components

$$v = \sqrt{v_x^2 + v_y^2} \quad (3)$$

3. horizontal and vertical displacement

$$x = v_0 t \cos(\theta) \quad (4)$$

$$y = v_0 t \sin(\theta) - \frac{1}{2}gt^2 \quad (5)$$

4. time of flight

$$t = \frac{v_0 \sin \theta + \sqrt{(v_0 \sin \theta)^2 + 2gy_0}}{g} \quad (6)$$

5. range

$$d = \frac{v_0 \cos(\theta)}{|g|} (\sqrt{(v_0 \sin \theta)^2 + 2gy_0}) \quad (7)$$

1.4 Interview

Interview with third party.(Malcolm, physics teacher)

What information should the tool show? e.g. maximum height, path of projectile, velocity vectors, units etc.

"Ideally, vector arrows that can show velocity, velocity components, height, range."

What parameters should you be able to adjust? e.g. launch angle, speed, height, air resistance, gravity etc.

"Exactly those."

How complex should the modelling be? Should it take into account air resistance, projectile shape etc.

"I don't think I'd have a use for projectile shape. Air resistance is nice to be able to include and possible bouncing."

How would you like to interact with the tool? e.g. pressing a button to launch, scroll through the timeline of the projectile (play it back) etc.

"Nice to be able to drag an initial velocity vector with the mouse, but sliders for angle and speed with optional keyboard input would be fine. Being able to scroll time is good."

What would you like to be able to create? e.g. add markers, draw lines, spawn projectiles in a locations/project from set locations.

"Launch positions, possibly barriers too, and a target that can be placed at any range or height."

How important is the visual design of the program: should it be 3D or 2D, and should it be simple or have a more realistic design?

"2-D is probably fine. Minimal design if it's to be used as a teaching tool. No distracting elements."

Would you rather have tools to measure specific distances and analyse the flightpaths of projectiles, or would you rather they simply be displayed while hovering?

"Don't mind."

For all of the above: which of these features would be most important for you?

"The basics being easy - so easily adjust launch speed and angle and easy to extract range and height data."

With this tool: PHeT Projectile Motion Tool, what do you like and what do you not like? How could it be improved?

"Love it. It's very good. Dunno."

Would it be useful to be able to save the current state of the program and return to it later?

"Not really."

Should the projectiles be able to collide, and should you be able to customise the shapes of the projectiles when colliding?

"Multi projectile? Sure. Not sure why I would customize the shape"

What other features would be useful for a Projectile Motion tool?

"Reversibility - set the launch speed and move the target and it will automatically adjust angle, or fixed angle and it will automatically adjust launch speed."

1.4.1 Resulting Objectives

1. The tool should include arrows that show the velocity and its horizontal and vertical components using initial velocity and angle for a projectile at any point in time
2. The tool should calculate and display the height of the projectile at any point and be able to automatically calculate the range
3. the tool should be able to use initial angle and velocity to display the motion of the projectile
4. the launch position of the projectile should be settable to any height, and there should be a target that can be set at any position, which will be able to register if the projectile hits it
5. the tool can have a minimal 2d design, focusing on displaying the earlier mentioned key information
6. the tool should be able to adjust the gravity

1.4.2 Extended Objectives (if i have time after completing initial objective)

1. the tool should have a mode that allows us to position the target and set the initial speed and launch height, or to set the initial angle and launch height. The tool should then automatically calculate the angle or initial speed
2. being able to add air resistance to the simulation and being able to adjust it
3. add barriers that stop the projectile
4. be able to set initial velocity and angle using click and drag with a mouse

1.5 Analysis of Existing Solutions

<https://phet.colorado.edu/en/simulations/projectile-motion>

<https://ophysics.com/k8.html>

1.6 Objectives

Numbered and sub-numbered. Clearly defined and small in scope e.g. "a button will do this", "when the object hits the ground this will happen".

Currently none past those obtained from interviewing the third party so refer to that

1.7 Modelling

Sketch proposed solution e.g. Sketch the GUI. (I sketched the GUI on a whiteboard but forgot to take a picture, will remake but not enough time before deadline for analysis)

2 Documented design

2.1 Overall flowchart

I created an overall flowchart to demonstrate how any properties of the projectile are obtained from inputs and also what they output which can then in turn be used to find other properties

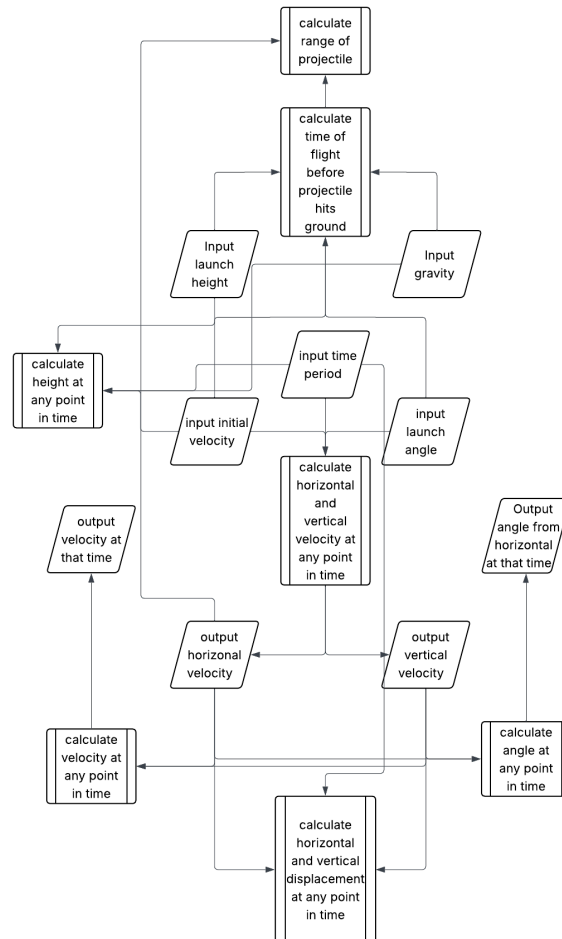


Figure 1: Overall design of flow in code

2.2 Hierarchy diagram

It would be ideal to create a UML diagram to represent how everything in the program would interact, however, before writing everything in terms of OOP I have initially created a diagram that effectively shows how all properties are inherited by the initial 5 user entered properties. If time allows before the deadline I'll create a proper UML version of this

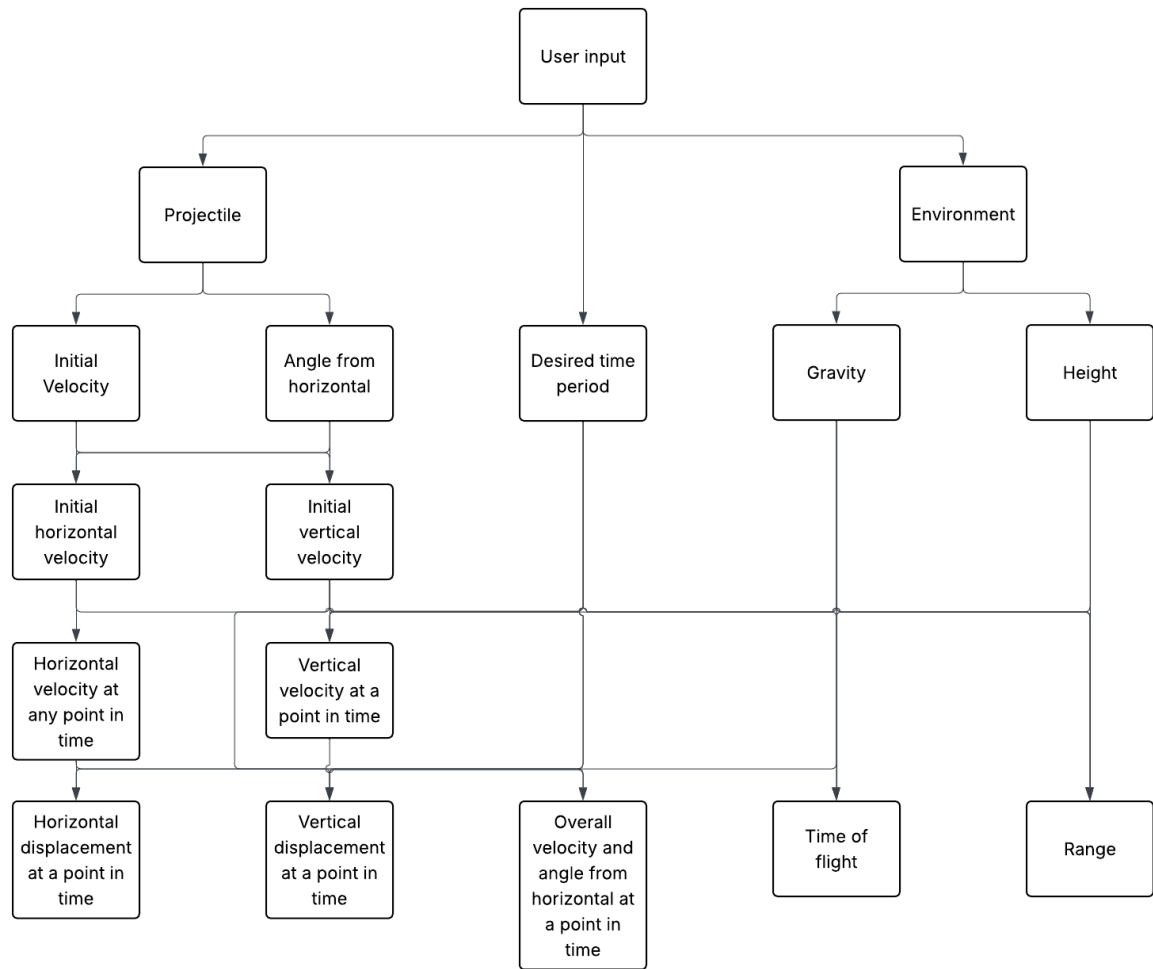


Figure 2: Inheritance diagram

3 Technical solution

4 Testing

5 Evaluation